

Closed-Loop feedback control system for EXOhSPEC

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ABSTRACT

EXOhSPEC is an ambient-air, fibre-fed, high-resolution spectrograph testbed that records optical-path-length metrology, image-centroid position and environmental telemetry during operation. We test a closed-loop architecture in which a thermo-electric cooler provides slow image-position correction and a discrete fine-optics stage provides bounded residual correction, and we measure each actuator’s individual contribution against an open-loop baseline recorded on the same hardware. Over a 31.44-hour hybrid run with both actuators enabled, the image position stayed within 0.5 pixels of the operating reference for all 1604 feedback frames, with a scatter of 0.0635 pixels horizontally and 0.1524 pixels vertically. Disabling the fine-optics stage and correcting with the thermal channel alone degraded this to 0.316 pixels. Disabling the thermal channel after warm-up and correcting with the fine-optics stage alone gave 0.138 pixels – matching the full hybrid result with an order of magnitude fewer corrective actions, which indicates that fine-scale stabilisation in this architecture is carried mainly by the fast actuator. Disabling both actuators entirely, immediately after an extended hybrid run on the same hardware, increased the image-position scatter eleven-fold: a direct, measured comparison rather than an estimate from separate sessions. The monitored optical path continued to evolve throughout every closed-loop record even while the image position remained bounded, so a controlled image position and a stationary optical path are separate outcomes. We report the control law, the identified environmental model and the safety logic in full, so the actuator roles described here can be checked directly against the matched open-loop and single-actuator comparisons.

Key words: instrumentation: spectrographs – techniques: radial velocities – optical metrology – control systems – atmospheric effects – methods: data analysis

1 INTRODUCTION

High-resolution radial-velocity spectrographs are stabilised as complete systems rather than by controlling a single temperature sensor or optical component. The measured spectrum responds to the thermal and mechanical state of the optical train, to pressure and refractivity along the beam, and to detector behaviour, illumination stability, wavelength calibration and extraction (Fischer et al. 2016). Since the first detections of exoplanets by the Doppler method (Mayor et al. 2003), the field has advanced by identifying each of these contributions and controlling it to a stated tolerance: HARPS and, more recently, ESPRESSO reach sub-ms^{-1} stability by combining vacuum enclosures, active thermal control and pressure regulation with careful fibre scrambling and calibration (Mayor et al. 2003; Pepe et al. 2021; Spronck et al. 2015). The literature therefore separates design requirements, subsystem validation and performance verification, and this separation frames the present work: our data measure image-centroid displacement and one monitored interferometric path, not a complete stellar radial-velocity error budget.

The following instrumentation studies inform our approach rather than serving as performance benchmarks an ambient-air testbed should match. The original EXOhSPEC instrument established a

compact, off-the-shelf, fibre-fed spectrograph with $R > 70,000$ in which thorium–argon line positions were monitored in real time and a piezo-actuated element removed image drift (Jones et al. 2021). Here we do not repeat that optical-design work; instead we ask whether optical-path monitoring, environmental telemetry, thermal correction and bounded fine-optics trim together form an interpretable closed loop, and whether each actuator’s contribution can be isolated and measured directly. The NEID environment-control programme converts an error-budget requirement into a documented control architecture and a long-duration stability demonstration in vacuum (Robertson et al. 2019); for our work this means the measured channels, time window, reference origin and actuator limits all need to be stated, and that ambient-air control should not be presented as equivalent to vacuum stabilisation. EXPRES verifies performance by decomposing the instrumental error budget into environmental, mechanical, optical, calibration, fibre and detector contributions (Blackman et al. 2020); for the same reason, every claim here is restricted to the directly measured quantity, and no radial-velocity number is inferred without a wavelength solution and extraction analysis. The NEID fibre-feed study also shows that a stable environmental trace alone does not eliminate illumination-driven error (Kanodia et al. 2023), so image position, optical path, actuator commands and environmental context need to be interpreted together. These requirements have been consolidated into comprehensive error budgets

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for next-generation Doppler spectrometers (Halverson et al. 2016), against which laboratory subsystem tests such as this one can be positioned.

The physical motivation for recording pressure, temperature and humidity is specific and quantitative. For a geometric path length L in air of refractive index n , the leading-order change in optical path is $\delta\text{OPL} \approx n \delta L + L \delta n$: a mechanical term and a refractivity term of comparable importance. The refractivity of air, $n - 1 \approx 2.7 \times 10^{-4}$ at optical wavelengths, varies at the level of roughly 1×10^{-6} per kelvin, 3×10^{-7} per hectopascal and 1×10^{-8} per per-cent relative humidity near room conditions (Ciddor 1996; Bönsch & Potulski 1998). Over a metre-scale air path a one-kelvin drift therefore changes the optical path by of order a micron, which is directly comparable to the interferometric residuals reported below and to the pixel-scale image motion of interest. Recording the atmospheric state synchronously with the optical path is thus not incidental telemetry but a prerequisite for interpreting the measured lengths at all; the same first-order relation, however, prevents a single centroid displacement from being assigned uniquely to a geometric or a refractive cause, because mechanical, thermal-boundary and illumination effects coexist. Figure 1 shows this coupling directly rather than only from the scaling argument above: a small step in the recorded temperature coincides with a distinct dip in the simultaneously monitored optical path, visible in both the raw and smoothed traces, over a 2-hour window in which humidity and pressure evolve smoothly and cannot account for the feature.

A closed-loop, or feedback, control system is one in which the quantity to be stabilised is measured, compared against a desired reference, and used to compute a corrective action that is applied back to the system, so that the effect of the correction is itself measured on the next cycle (Åström & Murray 2008). This differs fundamentally from open-loop or feed-forward operation, in which a correction is applied from a model or a schedule without measuring its result: an open-loop scheme cannot compensate for a disturbance it did not anticipate, nor detect when its own model has ceased to be valid. For a spectrograph the distinction is decisive. The disturbances that move the image on the detector—laboratory temperature cycling, barometric drift, slow mechanical relaxation—are neither fully predictable nor constant between nights, so a fixed corrective schedule computed once cannot hold the image in place indefinitely. A feedback loop instead measures the image position continuously and corrects only what it observes, which allows it both to reject unforeseen drift and, critically, to expose the moment its internal model no longer describes the instrument.

Implementing such a loop for EXOhSPEC requires three elements: an observable that reports the controlled quantity, one or more actuators that can move it, and a decision layer that maps the measured error to a bounded actuator command. Here the observable is the detector-plane centroid of a calibration feature, monitored alongside an independent optical-path length and the ambient state; the actuators are a slow thermoelectric channel that sets the enclosure thermal state and a fast fine-optics stage that trims the image position in discrete steps; and the decision layer identifies an environmental model from the current record and issues gated, magnitude-limited corrections. Sections 2 and 3 specify this architecture and its control law in full, including the calibration mathematics for the fine-optics stage. Section 4 then reports, against recorded data, whether the resulting loop holds the image position within a controlled region, and – the central addition of this revision – what happens when each actuator is disabled in turn and when all actuation is removed entirely on the same hardware.

The control-engineering context is equally specific. The plant here

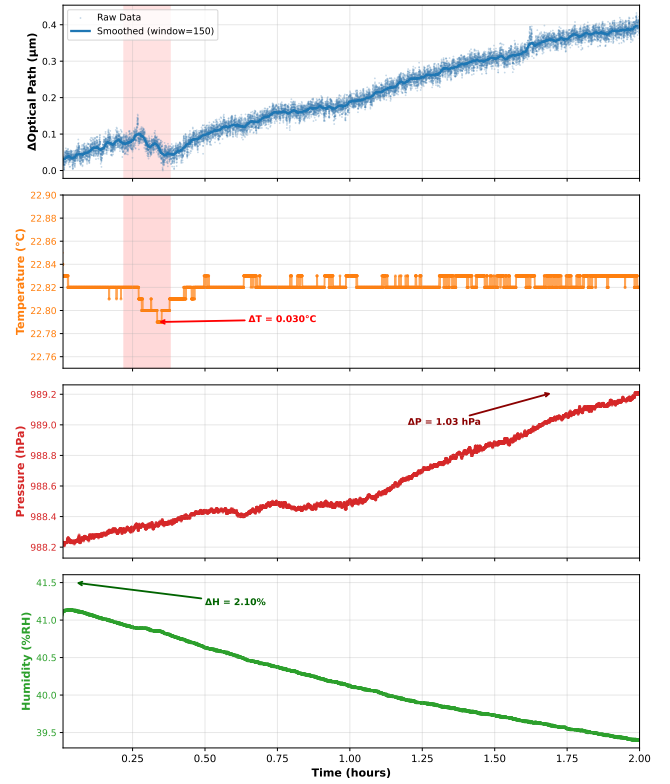


Figure 1. Synchronous optical-path and environmental telemetry over a 2-hour window. (a) Raw (light points) and smoothed (dark line, 150-sample moving average) optical-path length. The shaded region marks a localized excursion near $t \approx 0.3$ h. (b) Temperature, showing a $\Delta T = 0.030^\circ\text{C}$ dip coincident with the excursion in (a); pressure and humidity evolve smoothly over the same interval and show no matching feature, isolating temperature as the driver of this particular excursion. (c) Pressure, increasing by $\Delta P = 1.03$ hPa over the window. (d) Humidity, decreasing by $\Delta H = 2.10\%$ over the same period. Over the full window, the optical path’s longer-timescale rise tracks the pressure increase and humidity decrease, while the short-timescale excursion near $t \approx 0.3$ h tracks the temperature dip; together these two timescales motivate the synchronous environmental recording and closed-loop compensation described in Section 3.

combines a slow, nonlinear thermal channel with a fast but discrete actuator, a combination for which a single continuous regulator is poorly suited: continuous fine correction of a thermal setpoint invites oscillation, while a discrete actuator with finite travel cannot supply arbitrary continuous correction (Åström & Murray 2008). Model-based correction of such a plant further requires that the input–output relation be identified from the current record rather than transferred unchanged from an earlier state (Ljung 1999), a point returned to in Section 5. This paper asks: can a time-synchronised feedback loop, using a slow thermal channel and a bounded fast channel that re-identifies its environmental model on each run, hold the measured image position within a controlled region, and how much of that result comes from each actuator individually rather than from the pair acting together? We address this in three parts: the stability each actuator achieves alone, compared with both acting together; what a matched, same-hardware open-loop baseline shows the closed loop is worth; and which measurements are still needed before a stronger stability or radial-velocity claim is justified.

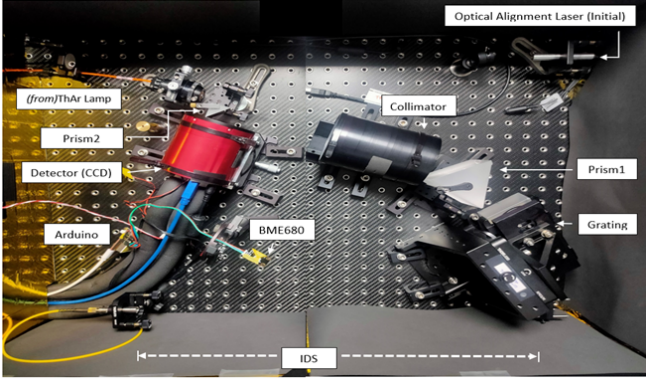


Figure 2. EXOhSPEC laboratory spectrograph testbed. The figure identifies the optical train, detector, environmental instrumentation and interferometric monitor used in the feedback experiments. It defines experimental context rather than a detailed optical prescription.

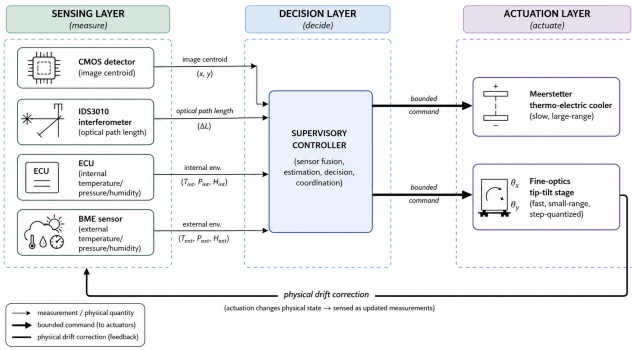


Figure 3. Signal-flow architecture of the closed loop: a sensing layer (detector centroid, interferometric optical path, internal and external environment), a decision layer (sensor fusion, estimation, decision and coordination), and an actuation layer (thermo-electric cooler, fine-optics stage), closed by the actuators’ physical effect on the next sensing-layer reading.

2 EXPERIMENTAL CONFIGURATION

2.1 EXOhSPEC testbed and available channels

EXOhSPEC is a fibre-fed laboratory echelle spectrograph operated at ambient pressure. A selected thorium–argon calibration feature is measured on a scientific CMOS detector. The feature centroid provides an image-plane observable. An IDS3010 interferometer monitors a fixed metrology path, while ECU and BME channels record temperature, pressure and relative humidity. A Meerstetter TEC controller changes the slow thermal state, and the fine-optics unit supplies discrete image-position commands. Figure 2 identifies the laboratory components at the level needed for the present experiment. Detailed optical alignment geometry and implementation-specific calibration matrices are not required to interpret the reported feedback evidence.

Figure 3 recasts the same components as a signal-flow architecture, in the three-layer form used throughout this section: a sensing layer (the four channels above), a decision layer (the supervisory controller of Section 3), and an actuation layer (the two actuators of Section 3), closed by the actuators’ physical effect on the next sensing-layer reading. This layering is common to every configuration in Section 4 and to the extended architecture described in Section 5; only the internal logic of the decision layer differs between them.

The measured and commanded quantities each carry a specific interpretation limit that is retained throughout the analysis. The image

position is the centroid (X, Y) of one selected thorium–argon feature on the CMOS detector; a single feature position is not, by itself, a wavelength solution or a radial-velocity observable. The reported closed-loop output is the displacement $(\Delta X, \Delta Y)$ and radial error r of that centroid relative to a declared origin, and a small scatter in these quantities does not prove that the origin itself was recovered. The optical path OPL and its change ΔOPL come from an independent IDS3010 metrology path, which monitors a fixed reference path and is not the sum of every image-forming path in the spectrograph. Temperature, pressure and humidity from the ECU and BME sensors provide environmental context and local-model inputs, but these point sensors do not map the full component-temperature field. The Meerstetter thermo-electric controller supplies the slow primary correction and a thermal-state record, noting that a commanded setpoint change is not an instantaneous plant response. The fine-optics unit supplies discrete image-position steps with a monitored cumulative state; being discrete and of finite travel, it cannot supply arbitrary continuous correction. These channels are recorded synchronously, so image position, optical path, actuator commands and environmental context are interpreted together instead of as single-variable causal explanations of image motion.

2.2 Image displacement, optical path and reference origin

At sample k , the selected feature has centroid (X_k, Y_k) . Relative to a stated origin (X_0, Y_0) , the image displacement is

$$\mathbf{d}_k = \begin{bmatrix} \Delta X_k \\ \Delta Y_k \end{bmatrix} = \begin{bmatrix} X_k - X_0 \\ Y_k - Y_0 \end{bmatrix}, \quad r_k = (\Delta X_k^2 + \Delta Y_k^2)^{1/2}. \quad (1)$$

The monitored optical-path change is written

$$\Delta\text{OPL}_k = \text{OPL}_k - \text{OPL}_0. \quad (2)$$

For geometric path length L in air with refractive index n , the leading-order perturbation is

$$\delta\text{OPL} \approx n \delta L + L \delta n. \quad (3)$$

Equation 3 motivates synchronous OPL and environmental recording. It does not allow the source of a particular feature-centroid displacement to be assigned uniquely to δn or to δL .

For any retained window of N samples, the scatter about that window’s mean and the root-mean-square distance from the fixed origin are reported separately:

$$\sigma(q) = \left[\frac{1}{N-1} \sum_{k=1}^N (q_k - \bar{q})^2 \right]^{1/2}, \quad \text{RMS}_0(q) = \left[\frac{1}{N} \sum_{k=1}^N q_k^2 \right]^{1/2}. \quad (4)$$

Section 4 reports four closed-loop configurations recorded on the same hardware – a hybrid baseline, two single-actuator variants that isolate each actuator’s contribution, and a matched open-loop baseline recorded on the same run as the hybrid result – so that Equation 4 can be applied directly across configurations, without a cross-run percentage correction.

2.3 Definition of the closed loop

Figure 4 shows the actual experimental hierarchy. The measured signals are image position, OPL and environmental telemetry. A supervisory decision layer receives these time-synchronised values and issues bounded commands. The TEC receives the slow correction role; the fine-optics stage is retained for residual image-plane correction with cumulative travel monitored. The diagram does not claim

a fully identified global model. It shows the measurement-to-action-to-measurement chain that makes the experiment a closed-loop test.

3 CONTROL ALGORITHM AND IDENTIFIED MODEL

The supervisory layer of Fig. 4 is an event-driven controller. Rather than transferring a fixed input–output relation between runs, it re-identifies its environmental model online from the current record (Ljung 1999) and drives the slow thermal channel through a gain-scheduled proportional–integral–derivative (PID) law, retaining the fast channel as bounded trim. This section states the model and the control law; Section 3.5 gives the safety logic that gates their application.

3.1 Environmental optical-path and pixel models

The optical-path residual is represented by a ridge-regularised linear model (Hoerl & Kennard 1970) in the incremental environmental variables — changes in temperature, pressure, absolute humidity and an internal-temperature proxy, with an outdoor-temperature term added in the later configuration described in Section 4.3,

$$\widehat{\Delta\text{OPL}} = g_T \delta T + g_P \delta P + g_H \delta \text{H}_2\text{O} + g_{T_{\text{in}}} \delta T_{\text{in}} + g_{T_{\text{out}}} \delta T_{\text{out}} + c, \quad (5)$$

whose coefficients are obtained over a rolling window of N_w recent samples from the regularised normal equations

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}, \quad (6)$$

with design matrix $\mathbf{X}$ carrying the five predictors and an intercept, and a small ridge parameter λ that stabilises the inversion when predictors are collinear, as temperature and refractivity necessarily are. Because temperature is known a priori to be the dominant optical-path driver, an updated temperature coefficient is accepted only if it exceeds a floor in magnitude and preserves the sign established during warm-up; a vanishing or sign-reversed g_T is treated as an ill-conditioned fit and the previous value is retained. The fit quality is recorded per update as the coefficient of determination

$$R^2 = 1 - \frac{\sum_k (y_k - \hat{y}_k)^2}{\sum_k (y_k - \bar{y})^2}, \quad (7)$$

which is used both as a diagnostic and as a gate on the feedforward path (Section 3.5). A parallel ridge model maps the same environmental increments to the pixel displacements $(\Delta X, \Delta Y)$, subject to an analogous sign-consistency test on the vertical coefficient, so that the reactive and feedforward corrections cannot be driven by a spurious fit.

3.2 Fused thermal control law

The two model channels are combined into a single temperature-equivalent error with a fixed weighting that favours the direct optical-path estimate,

$$e_T = w e_T^{\text{OPL}} + (1 - w) e_T^{\text{px}}, \quad w = 0.80, \quad (8)$$

and the slow-channel effort is the gain-scheduled PID response to this error augmented by feedforward and a fine-optics load-absorption term,

$$\Delta u = \kappa_b \left(k_p e_T + k_i \int e_T dt + k_d \dot{e}_T \right) + e_T^{\text{ao}} + K_{\text{ff}}^{\text{OPL}} \widehat{\Delta T}^{\text{OPL}} + K_{\text{ff}}^{\text{px}} \widehat{\Delta T}^{\text{px}}. \quad (9)$$

Table 1. Error bands, PID gain multipliers and asymmetric heat/cool step caps used by the gain-scheduled thermal law. Bands are defined on the radial image error r ; caps are per-cycle limits on the commanded temperature increment.

Band (r range, px)	(g_p, g_i, g_d)	heat ($^{\circ}\text{C}$)	cool ($^{\circ}\text{C}$)	settled
< 0.05	(0.25, 0.20, 0.50)	0.005	0.003	12
0.05–0.20	(0.40, 0.40, 0.35)	0.015	0.010	8
0.20–0.60	(0.75, 0.75, 0.65)	0.035	0.020	6
0.60–1.20	(1.00, 1.00, 1.00)	0.080	0.050	4
1.20–2.50	(1.10, 1.00, 1.05)	0.160	0.100	3
≥ 2.50	(1.15, 1.00, 1.05)	0.250	0.160	3

Here κ_b is the band multiplier of Table 1, the integral term is clamped for anti-windup and accumulated only while the gates are open, and the feedforward gains are engaged only once the model quality of Equation 7 is adequate. The commanded increment then passes a fixed cascade before it reaches the hardware: a deadband that suppresses corrections below the measurement noise, the band-dependent asymmetric step cap, a minimum step, a rate limit, and a sign-flip suppressor that blocks an immediate reversal. This cascade converts the analytically optimal effort into a physically admissible, non-oscillatory setpoint change appropriate to a slow thermal actuator.

The error-classification bands and their asymmetric heat/cool step limits are given in Table 1. Authority is made proportional to error: near the target the controller applies gentle, tightly capped corrections to avoid injecting noise, while at large error it is permitted larger and faster corrections. The heat and cool caps differ because the thermal plant heats and cools at different effective rates.

3.3 Bounded fast-channel correction

The fine-optics stage corrects residual image error through a calibrated linear map between its two raw step axes (s_x, s_y) and detector-plane motion,

$$\begin{pmatrix} s_x \\ s_y \end{pmatrix} = \mathbf{M}^{-1} \begin{pmatrix} -\Delta X \\ -\Delta Y \end{pmatrix}, \quad \mathbf{M} = \begin{pmatrix} 0.150 & -0.045 \\ 0.064 & 0.152 \end{pmatrix}, \quad (10)$$

with the result rounded to integers and clamped to a single step per cycle. A single step corresponds to ≈ 0.16 px of image motion, which sets a quantisation scale for the fast channel: because this step can exceed a small residual, a correction may overshoot and require a reverse step on a later cycle, producing the discrete back-and-forth pattern characteristic of a bounded relay actuator. A sensitivity analysis over the recorded data shows that the vertical residual is not limited by this quantisation — removing the quantisation variance ($\text{step}/\sqrt{12} \approx 0.046$ px) from the measured $\sigma(\Delta Y)$ leaves ≈ 0.145 px — so a finer fast stage would reduce chatter but not the residual floor, which is set by grating-mount thermal motion and optical-path drift. Adopting a finer stage requires only substituting the numerical entries of $\mathbf{M}$; the surrounding algorithm is unchanged, provided the replacement passes the calibration and validity check set out in Section 3.4. The cumulative fast-channel state is tracked as a ledger and gently unloaded toward centre when it approaches its travel limit, with the released load transferred to the slow channel, so that the fast actuator always retains authority for fine correction and never becomes the dominant long-term corrector.

3.4 Calibration and validity check for the fast-channel matrix

Equation 10 is not assumed; it is measured, and the measurement carries its own validity check before it is accepted into the control

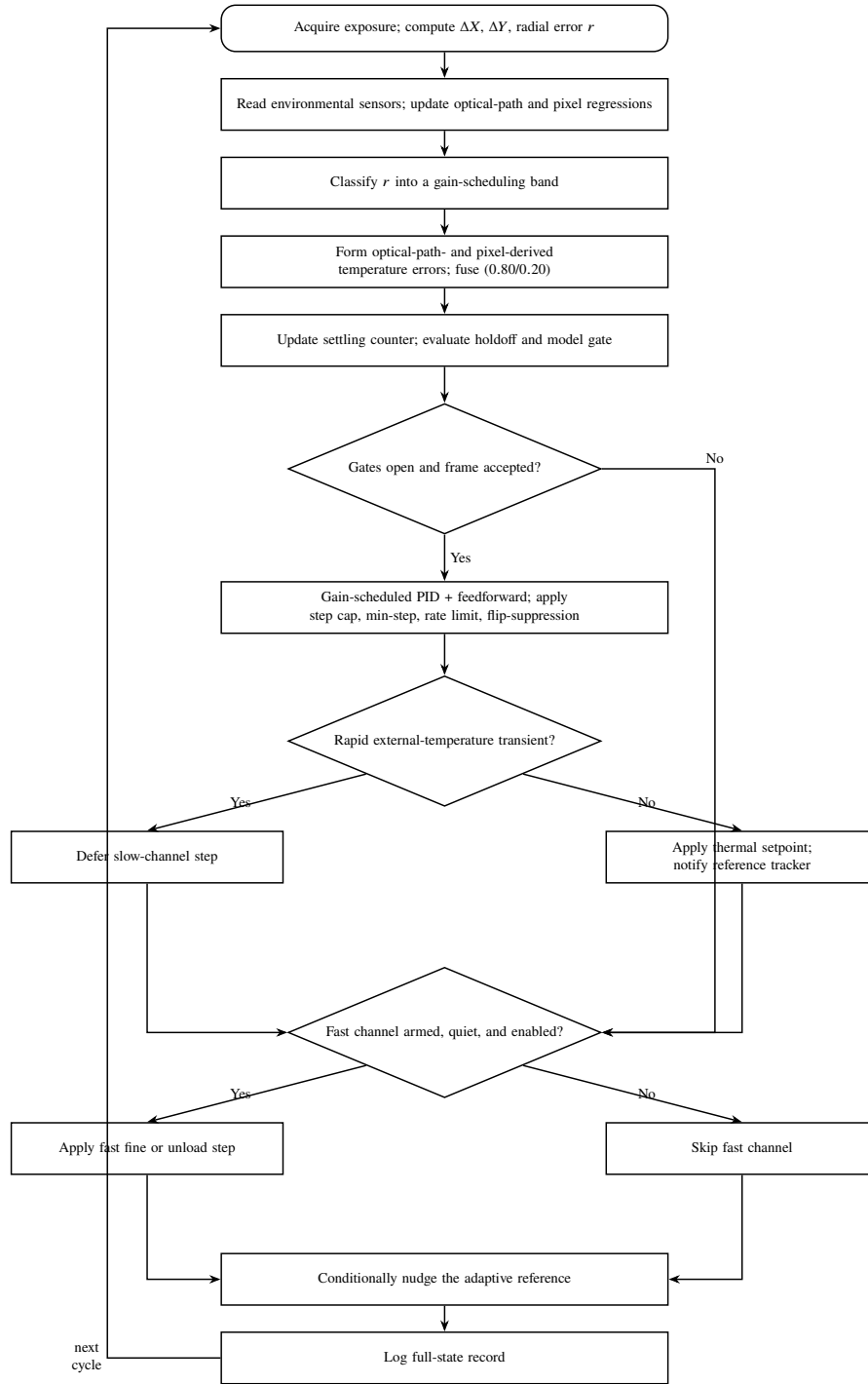


Figure 4. Per-cycle decision flow of the closed loop. Each exposure yields an image displacement and radial error; the environmental model is updated, a gain-scheduling band is selected, and a fused reactive/feedforward temperature error drives the gain-scheduled thermal correction subject to settling, holdoff, model-quality and external-transient gates. The fast fine-optics channel is applied only as bounded trim when armed, quiet and enabled, and an adaptive reference absorbs slow residual bias under strict stability conditions. This is the architecture common to every configuration in Section 4; the single-actuator variants disable one branch of this diagram while leaving the rest, including all safety gates, unchanged.

law. The calibration protocol commands a sequence of step counts $n \in \{1, 10^2, 10^3\}$ along each of the four cardinal actuator directions (north, south, east, west), re-seeding the measured centroid between phases and separating repeats by a fixed cooldown so that consecutive commands do not contaminate one another. For each commanded step vector $\mathbf{s}_j$ the resulting centroid displacement $\mathbf{d}_j = (\Delta X_j, \Delta Y_j)$

is recorded, and the two independent columns of $\mathbf{M}$ are obtained from the least-squares slope of displacement against commanded

step count along each nominal axis,

$$\hat{\mathbf{m}}_a = \arg \min_{\mathbf{m}} \sum_{j \in a} \|\mathbf{d}_j - n_j \mathbf{m}\|^2 = \frac{\sum_{j \in a} n_j \mathbf{d}_j}{\sum_{j \in a} n_j^2}, \quad a \in \{\text{N-S, E-W}\}, \quad (11)$$

with $\mathbf{M} = [\hat{\mathbf{m}}_{\text{E-W}} \ \hat{\mathbf{m}}_{\text{N-S}}]$. The 1-step phase is retained only as a noise reference: at that step count the displacement is comparable to the centroid-measurement scatter itself and is excluded from the fit, whereas the 10^3 -step phase, at roughly thirty times the single-step displacement, dominates Equation 11 and sets the reported calibration.

Because the two actuator axes are commanded independently, a physically sound calibration is expected to satisfy three checks before Equation 10 is used operationally. First, opposite-direction commands should give displacement magnitudes that agree to within measurement noise, $|\hat{\mathbf{m}}_{\text{N}}| \approx |\hat{\mathbf{m}}_{\text{S}}|$ and $|\hat{\mathbf{m}}_{\text{E}}| \approx |\hat{\mathbf{m}}_{\text{W}}|$. Second, the two nominal axes should be close to orthogonal in the detector frame, quantified by the angular deviation

$$\theta_{\text{err}} = \left| \arccos \left(\frac{\hat{\mathbf{m}}_{\text{N-S}} \cdot \hat{\mathbf{m}}_{\text{E-W}}}{|\hat{\mathbf{m}}_{\text{N-S}}| |\hat{\mathbf{m}}_{\text{E-W}}|} \right) - 90^\circ \right|, \quad (12)$$

with a working acceptance threshold of $\theta_{\text{err}} < 5^\circ$ per axis. Third, each commanded direction should produce a measured displacement within a few degrees of its own expected bearing; a measured bearing displaced by approximately 180° from the commanded one indicates a reversed axis mapping in the actuator wiring or firmware rather than a calibration or alignment error, since a wiring or firmware reversal reproduces the correct displacement magnitude and the correct orthogonality between axes while inverting only the sign of one axis.

Applying this protocol to a candidate finer-step actuator intended as a drop-in replacement for Equation 10 demonstrated the check acting as intended. The north–south axis passed both checks ($\theta_{\text{err}} \approx 4\text{--}5^\circ$, opposite-direction magnitudes agreeing to within a few per cent), while the east–west axis initially showed the 180° bearing-reversal signature above – commanding east produced a westward displacement and vice versa, with the correct magnitude and the correct 90° separation from the north–south axis once the sign was accounted for. The fault was traced to a command-mapping swap in the actuator firmware rather than a hardware failure, and was corrected at the firmware level. A repeat 10^3 -step verification after the correction found all four cardinal directions within tolerance of their expected bearing (angular error $2\text{--}6^\circ$, against a $5\text{--}15^\circ$ per-axis acceptance band; two directions passed with margin, two returned a marginal warning inside the band), with the step size unchanged from the pre-fault calibration. The check is reported here because it is itself a result: the same validation approach that governs the thermal control law (Section 3.5) is applied to actuator calibration, and it caught a firmware-level axis reversal before that fault could reach a closed-loop run.

3.5 Safety and validity gates

Corrections are applied only when the state is quiescent and the model is trustworthy, which is enforced by a set of gates that all reduce to withholding action when the assumptions behind Equations 5–9 are not met. A thermal step requires both a minimum holdoff since the previous step and a band-dependent settling count (Table 1), where the settling counter increments only while the optical path and vertical image error are simultaneously calm and resets on disturbance; this prevents the controller from acting on transient noise or chasing a moving disturbance. The fast channel is held off for a fixed interval after any thermal action, relaxed after a small step, because a small

setpoint change settles within a few frames and a full holdoff would suppress fine correction precisely when it is most effective. Thermal action is deferred during rapid external-temperature transients, identified from a least-squares slope of the external temperature, because during such a swing the environmental model is transiently unrepresentative and a correction computed then would be applied against a disturbance about to reverse. For an initial interval after warm-up the loop runs with the fast channel disabled so that the model stabilises before the fast actuator engages, and the feedforward terms of Equation 9 are ramped in only once the model quality of Equation 7 exceeds a threshold. Frames with pixel outliers are excluded from the regression, and a frame flagged as unreliable forces zero effort and a hold state; when the model cannot be trusted the controller falls back to slow-channel-only operation. Finally, before any exposure the detector cooler is commanded and the loop waits until the measured sensor temperature has settled, because the cooler’s thermal load couples into the optical path during warm-up and otherwise corrupts the warm-up model seed.

The single-actuator configurations of Section 4 are produced by disabling one branch of Fig. 4 at the configuration level – the fast channel switched off entirely, or the thermal channel frozen at its warm-up setpoint after the warm-up phase completes – while every gate above remains active on whichever channel is still enabled. The open-loop configuration disables both branches after an otherwise-normal hybrid run completes, with logging continuing unchanged.

The spectral method used to characterise the residual optical-path variation in Section 4.2 is the averaged-periodogram method of Welch (1967), applied to the linearly detrended optical-path record. The frame cadence of approximately 70 s sets a Nyquist limit near 7×10^{-3} Hz, so the method resolves periodicities from a few minutes to hours and is not sensitive to acoustic or mechanical vibration in the hertz-to-kilohertz range, which would require dedicated high-rate sensing.

4 RESULTS

4.1 Actuator decomposition and a matched open-loop baseline

Four closed-loop configurations of the architecture in Sections 2–3 were run on the same hardware: a hybrid run with both actuators enabled (31.44 h feedback, 1604 frames); a thermal-channel-only run with the fast stage disabled (2.95 h feedback, 152 frames); a fast-channel-only run with the thermal channel frozen at its warm-up setpoint (4.06 h feedback, 208 frames); and an extended hybrid run (32.24 h) immediately followed, on the same session, by an open-loop interval (23.31 h) in which both actuators were disabled entirely. Table 2 reports the feedback-phase statistics for all four; Fig. 5 shows the annotated $\Delta X/\Delta Y$ record for each.

The thermal channel alone is the weakest of the three active configurations (0.3164 px, 88.8 per cent within 0.5 px), despite issuing corrections throughout the run. The fast channel alone matches the full hybrid result almost exactly (0.1378 px against 0.1524 px) while issuing 13 commands over 4.06 h, an order of magnitude fewer actions per unit time than the thermal channel manages on its own. Read together, these two results indicate that fine-scale image stabilisation in this architecture is carried disproportionately by the fast actuator, while the thermal channel’s role is closer to holding a coarse operating point than to suppressing sub-pixel oscillation – consistent with the two actuators’ designed bandwidths (thermal: slow, large range; fast optics: rapid, small range), but not something the architecture description in Section 3 establishes on its own. Both single-actuator

Table 2. Feedback-phase statistics across the four configurations, all recorded on the same hardware. Actuator command counts are per feedback phase; the open-loop row has zero by construction. Dashes denote a quantity not reported for that configuration.

Configuration	Duration (h)	Frames	$\sigma(\Delta Y)$ (px)	$\sigma(\Delta X)$ (px)	Within ± 0.5 px (%)	TEC cmds	Fast-channel cmds
Hybrid (both actuators)	31.44	1604	0.1524	0.0635	100.0	68	69
Thermal channel only	2.95	152	0.3164	0.0287	88.8	5	0
Fast channel only	4.06	208	0.1378	0.0602	100.0	0	13
Hybrid (extended run)	32.24	1996	0.1214	–	99.9	121	180
Open-loop (both disabled)	23.31	106	1.3565	–	59.4	0	0

runs are short relative to the hybrid record (2.95 h and 4.06 h against 31.44 h) and their statistics should be read as indicative rather than definitive, particularly in the tail percentiles.

The open-loop interval is more significant. Recorded on the same hardware, immediately following an extended hybrid run rather than compared against a historical or separately-scheduled baseline, disabling both actuators degraded $\sigma(\Delta Y)$ from 0.1214 px to 1.3565 px – an eleven-fold increase – and reduced the fraction of frames within 0.5 px from 99.9 to 59.4 per cent. Figure 6 shows the same result as a distribution rather than a summary statistic: the open-loop tail extends to -2.5 px, while the hybrid distribution remains concentrated within a few tenths of a pixel of zero, and Fig. 7 places all four configurations of Table 2 on the same axis for direct comparison. This is the first measurement in this programme of the closed loop’s benefit against a zero-actuation condition on the same session, rather than an estimate inferred from separate runs under different conditions.

4.2 Optical path during closed-loop operation

As in every configuration of Table 2, the monitored optical path continued to evolve during the hybrid run while the image position remained bounded: the OPL residual RMS was $0.4341 \mu\text{m}$ and the environmental model of Equation 5 explained approximately half of the OPL variance across the run (median $R^2 \approx 0.55$, interquartile range $[0.39, 0.68]$). Applying the averaged-periodogram method of Section 3.5 to this record places the dominant spectral feature at a period near 30 min, close to the mean interval between thermal-correction commands, with weaker features at shorter periods; a quantisation-floor estimate confirms the measured noise floor lies several orders of magnitude above the interferometer digitisation limit, so the residual reflects physical and environmental variation rather than measurement digitisation. This periodicity is reported as a correlation, not an established causal link, because separating an actuation-driven component from an environmental one requires a matched control-disabled record with optical-path monitoring retained throughout – a refinement of the open-loop result in Section 4.1, which is discussed as future work in Section 6. The dependence between a controlled image position and a still-evolving optical path is consistent across every hybrid record examined and is not specific to one run.

4.3 Does an added outdoor-temperature predictor improve the optical-path model?

The extended-hybrid configuration of Table 2 adds outdoor temperature, $g_{T_{\text{out}}}$ in Equation 5, as a fifth predictor, motivated by an earlier observation that outdoor temperature correlated with late-run vertical drift; the design target was to raise the median R^2 of Equation 7 toward 0.80–0.90. The result does not support that target: median $R^2 = 0.497$ for this configuration, close to the hybrid run’s own 0.546, and the $g_{T_{\text{out}}}$ coefficient stayed near zero throughout, indicating negligible outdoor-temperature sensitivity once internal

temperature, pressure and humidity were already accounted for. This result is nonetheless useful: it rules out the added predictor as a source of near-term R^2 improvement on this hardware. It also suggests, tentatively, that residual instability in earlier configurations owes more to actuator saturation than to the optical-path-model fit; we have not tested that explanation directly here, and do so in future work (Section 6). Because the extended-hybrid run may not have had much outdoor-temperature variation to fit against, this result is run-specific and does not rule out outdoor-temperature coupling in general; a longer run, or one with a larger outdoor-temperature swing, is needed before the predictor is dropped from the model.

5 DISCUSSION

Sections 3 and 4.1 together show two things: the environmental model is re-identified on each run rather than transferred between them, and the actuator roles reported here are supported by direct measurement, not by design intent alone. On re-identification specifically, an earlier stage of this programme carried a fixed control representation over from a previous run without re-identifying it; the run began with a stable interval of radial RMS ≈ 0.40 px but broke away to a full-record radial RMS of 5.892 px once the optical and environmental state had evolved past what the transferred representation described. The most plausible reading is that the controller was responding to a genuinely evolving disturbance with a model that was no longer valid, not that a single corrective action itself created the drift; the case is retained here only as the direct motivation for re-identifying the environmental model on every run, which every configuration in Section 4 does.

The actuator-decomposition result answers, with a direct measurement rather than an inference from the architecture’s design intent, which channel is doing the work. The fast-optics-only configuration matching the full hybrid result at one-tenth the command rate of the thermal-only configuration is a specific, falsifiable claim about this hardware, not a general statement about thermo-electric versus fast-optics correction: the thermal channel’s bandwidth and the fast stage’s step size were both fixed by the hardware available, and a different pairing of actuator bandwidths could shift the balance. Both single-actuator runs are short (2.95 h and 4.06 h) relative to the multi-day hybrid records, so the comparison should be treated as indicative of where the fine-stabilisation work is done on this system, not as a precisely bounded ratio.

Of the two new comparisons, the open-loop result matters more. An eleven-fold increase in ΔY scatter, measured on the same hardware immediately after the hybrid phase rather than inferred from a separate session, is direct evidence that the closed loop is doing something, in a way the earlier draft of this work could only assert. It does not, by itself, establish which part of the architecture – the thermal channel, the fast channel, the re-identified model, or their combination – is responsible for which fraction of that improvement; the single-actuator results of Section 4.1 address that question only



Figure 5. Annotated $\Delta X/\Delta Y$ records for all four configurations of Table 2, same axis scale throughout. Circles mark thermal-channel commands, triangles mark fast-channel commands. The thermal-channel-only record (b) shows visibly larger sustained oscillation than either configuration with the fast channel enabled; the open-loop segment of (d) is the only interval in which the trace leaves the ± 0.5 px band for an extended period.

partially, given their shorter duration. Nor does the open-loop comparison bear on the optical-path result of Section 4.2: the monitored optical path continued to evolve during every hybrid record examined, so a controlled image position remains a distinct outcome from a stationary optical path, and the environmental model of Equation 5 explained only about half of the OPL variance even in the best-conditioned run. For the same reason a detector image displace-

ment is not converted into a radial-velocity uncertainty here: that conversion requires a wavelength solution, local dispersion, line-spread-function behaviour, extraction stability, calibration fidelity and illumination analysis, and in the cited precision-radial-velocity literature a performance number is credible only when those contributing terms and their calibration paths are defined (Bouchy et al.

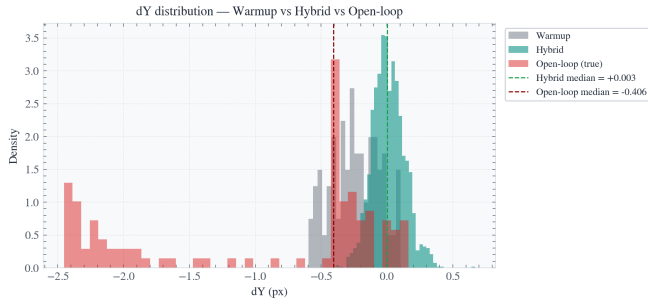


Figure 6. Distribution of ΔY for the extended-hybrid record of Table 2, split by warm-up, hybrid feedback and the subsequent open-loop interval. The open-loop distribution is offset and long-tailed (median -0.406 px, extending to -2.5 px); the hybrid distribution is centred near zero (median $+0.003$ px) and visibly narrower.

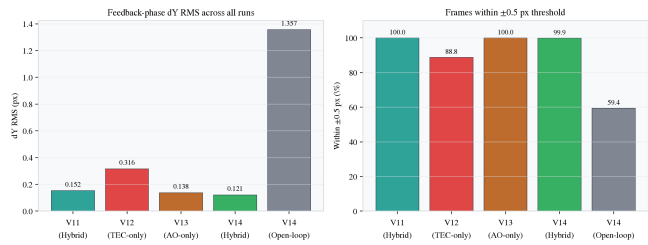


Figure 7. Feedback-phase ΔY RMS (left) and percentage of frames within ± 0.5 px (right) across all four configurations of Table 2, open-loop shown alongside for scale. The three active configurations cluster within a factor of ~ 2.6 of each other; open-loop is an order of magnitude worse than any of them.

2001; Fischer et al. 2016; Blackman et al. 2020; Halverson et al. 2016).

A separate, exploratory measurement supports the same actuator picture from a different angle. A localised thermal-impulse campaign, not itself closed-loop evidence, found that heating applied at the grating mount produced a measurable detector-plane image response, while heating applied near the camera mount coupled into both image motion and the monitored optical path; an enclosure or air-temperature sensor does not represent the temperature at either specific location, which motivates the component-level temperature sensing listed as future work below. A quantitative time-constant estimate was not attempted here because it requires a controlled step or impulse with a fitted exponential response, rather than the qualitative sensitivity mapping performed.

The same validation approach used for the thermal control law extends to actuator calibration itself: Section 3.4 describes a firmware-level axis reversal in a candidate fast-channel replacement, caught by the orthogonality and bearing checks before it could reach a closed-loop run, corrected, and reverified. Two of the four cardinal directions passed the post-correction check with margin and two returned a marginal warning inside the acceptance band, which we report as partially resolved rather than fully closed; a further verification pass is listed in Section 6.

This demonstrates, on an ambient-air testbed built from off-the-shelf components (Jones et al. 2021), that a dual-bandwidth controller which re-identifies its own environmental model can hold a spectrograph image position within a sub-pixel region over a multi-day interval without a vacuum enclosure, and that the actuator allocation and the closed loop’s net benefit can both be measured directly on

the same hardware. The same approach – disable one actuator, then disable both, on the same session – applies directly to other compact or cost-constrained spectrographs where full vacuum stabilisation is impractical, as a low-cost way of testing whether a given actuator’s added complexity is justified before committing to a more elaborate design.

5.1 An extended architecture under validation

An extended architecture, currently undergoing a multi-day validation run (Section 6), adds two mechanisms beyond those reported above. Neither altered the actuator-decomposition or open-loop results of Section 4, which used the architecture exactly as described in Sections 2–3; both are reported here as architecture, not as results, since the extended run has not yet completed.

The first is a fast correction lane for the fine-optics stage, summarised in Figure 8. The stage is armed once the radial error falls to a threshold and disarmed only once it rises past a second, higher threshold, a hysteresis band that prevents rapid re-arming on noise. While armed, it corrects on a longer cadence by default, but switches to a short, fast cadence specifically when the error is already small and the thermal channel’s own settle-gate (Section 3.5) has not yet released it to move – filling the window the slow channel is required to sit idle in, without altering the slow channel’s settle logic itself. A cumulative-travel check, independent of the arm state, unloads the stage in bounded bursts toward centre if it approaches its travel limit, with the thermal channel absorbing part of the released correction. Figure 9 illustrates the fast lane against the settle-gate directly: across an eight-frame window in which the thermal channel is held by its settle-gate, the fast lane fires three corrections rather than sitting idle.

The second is connection-health monitoring across the four measured channels (Section 2), summarised in Figure 10. A single channel failing for a sustained number of consecutive reads degrades the loop to a reduced-authority mode that continues safe operation on the remaining channels; two or more channels failing simultaneously degrades further, to a fully passive mode with no closed-loop updates. Both degrade automatically and recover automatically once the affected channel or channels return, and re-entry to full closed-loop operation never re-freezes or resets the environmental model, so a transient disconnection does not repeat the warm-up cost described in Section 2.

6 PLANNED VALIDATION EXPERIMENTS

Several validation experiments remain in preparation. Two items identified in an earlier stage of this work – a matched open-loop baseline and a slow-channel-only comparison – are now reported directly in Section 4.1 rather than remaining open. A defined set of further experiments is still required before a stronger stability or radial-velocity statement is justified. An independent repeat of the hybrid and open-loop configurations, under fresh environmental conditions and against acceptance criteria agreed in advance, will establish that the reported bounded operation and eleven-fold open-loop degradation are reproducible rather than specific to one session. A longer single-actuator comparison, at durations comparable to the multi-day hybrid record rather than the 2.95 h and 4.06 h runs reported here, will tighten the actuator-allocation result beyond an indicative one. A delayed thermal-response measurement – a small signed perturbation applied from an equilibrated, fixed reference at the grating and camera mounts specifically, with the delayed image and optical-path response fitted to a time constant – will convert the localised

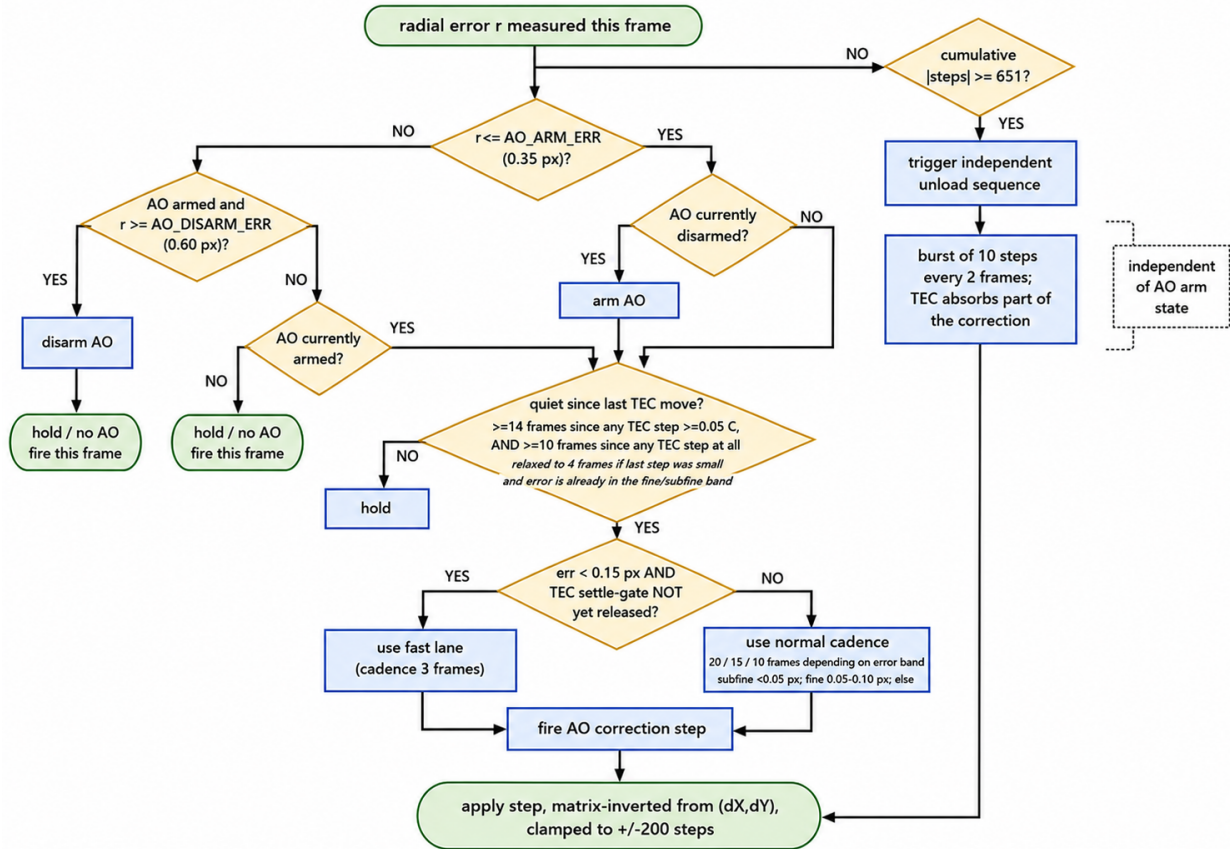


Figure 8. Arm/disarm and firing logic for the fast-optics correction stage in the extended architecture. Arming and disarming use separate thresholds (a hysteresis band); an independently-triggered unload sequence recentres the stage when its cumulative travel approaches its limit, regardless of arm state.

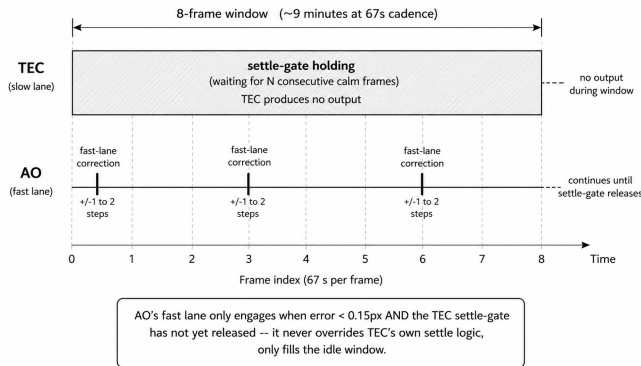


Figure 9. The fast lane against the thermal channel's settle-gate, over an eight-frame window at the nominal frame cadence. The fast lane engages only while the settle-gate holds and the error is already small; it fills the thermal channel's idle window rather than acting on its behalf.

sensitivity finding of Section 5 into a quantitative relation before it is used for correction. An open-loop record with optical-path monitoring retained throughout, rather than the actuator-only open-loop comparison reported here, is required to test whether the periodicity identified in Section 4.2 is actuation-driven or environmental. Finally, a further orthogonality verification of the fast-channel calibration in Section 3.4, targeting the two directions that returned a marginal warning after the firmware correction, will close that result out fully

rather than leaving it caveated. Together these convert the present bounded-feedback and actuator-decomposition demonstration into a reproducible, quantitative control model with defined uncertainty.

7 CONCLUSIONS

This paper reports a closed-loop experiment for the ambient-air EX-OhSPEC spectrograph, measuring image-centroid displacement and one monitored interferometric path under a dual-bandwidth controller that re-identifies its own environmental model on each run. New in this revision, we also decompose that result by actuator and compare it against a matched open-loop baseline recorded on the same hardware. The hybrid configuration held the image position within 0.5 px of the operating reference for all 1604 feedback frames of a 31.44 h run, with $\sigma(\Delta X) = 0.0635$ px and $\sigma(\Delta Y) = 0.1524$ px. Disabling the fast channel degraded this to 0.3164 px; disabling the thermal channel after warm-up left it effectively unchanged at 0.1378 px, using an order of magnitude fewer commands – so the fast actuator carries most of the fine-stabilisation work in this architecture. Disabling both actuators, immediately after an extended hybrid run on the same hardware, degraded $\sigma(\Delta Y)$ by a factor of eleven, giving a direct measurement of the closed loop's benefit rather than an inferred one. The monitored optical path continued to evolve throughout every hybrid record, and the environmental model explained only about half of its variance, so the result demonstrated here is bounded closed-loop image-position control – not recovery of a stationary optical path or a complete environmental model –

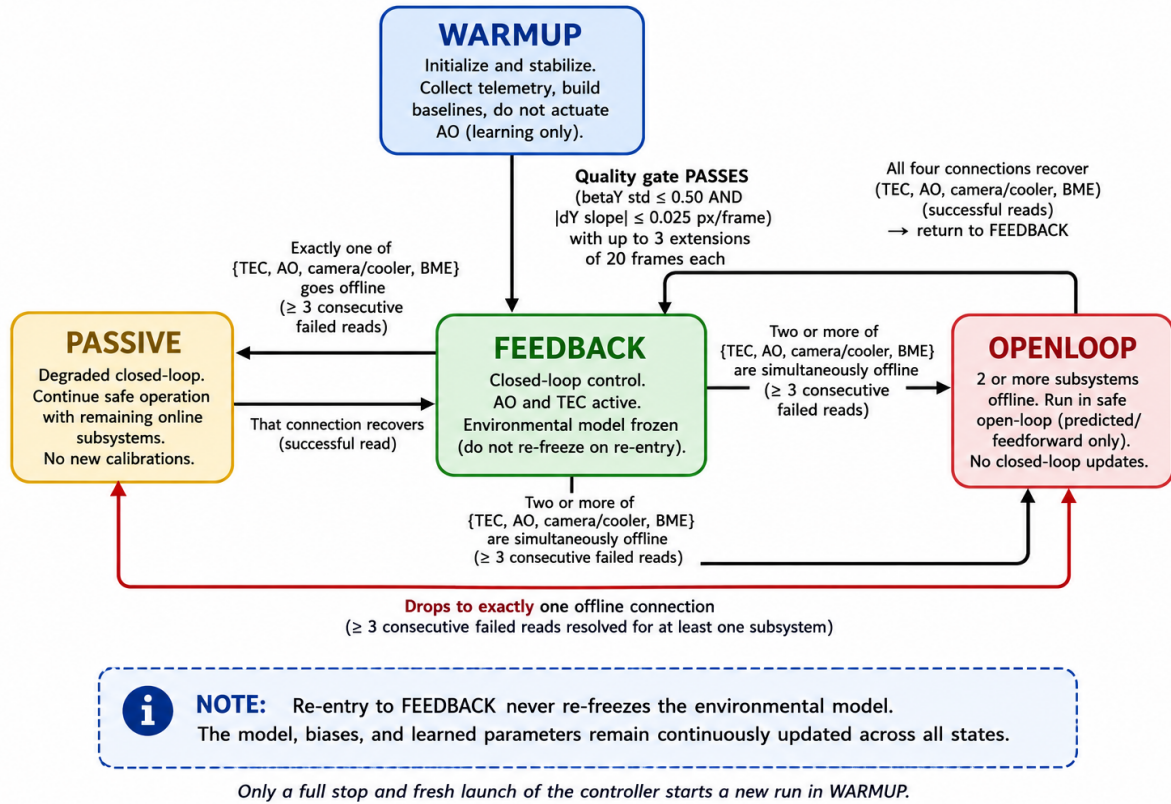


Figure 10. Connection-health state machine for the extended architecture. One failed channel degrades to a reduced-authority mode; two or more failed channels degrade to a fully passive mode; both recover automatically as connectivity returns, without re-freezing the environmental model.

and we do not infer a radial-velocity precision from it. The actuator-decomposition method itself – disable one channel, then both, on the same session – can be applied directly to other compact spectrographs to establish what a given actuator is contributing; the next tests needed are an independent repeat, a longer single-actuator comparison, and an equilibrium-referenced delayed-response measurement, set out in Section 6.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY

The reported summary metrics, figure-level products and analysis descriptions are retained in the EXOhSPEC project record. Raw laboratory telemetry, detector frames, detailed optical geometry, hardware settings and implementation-specific calibration files are not publicly released because they form part of an active instrument-development programme. Material required for scholarly assessment can be considered by the corresponding author subject to laboratory and collaboration approval.

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